

UCS5204 - Foundations of Artificial Intelligence — Lab

Ex:10 Case Study

6 nimmt! Game

Team Members

Keshav K S – 3122235001067

Sakthivel T – 3122235001111

1. 6 nimmt! – Game Description

Goal of the Game

Avoid collecting cards with bull points. The player with the fewest bull points at the end wins.

Game Materials

- **Deck:** 104 cards numbered 1–104.
- **Bull Points System:**
 - Cards ending in 5 → 2 points
 - Cards ending in 0 → 3 points
 - Double numbers (11, 22, 33, ... 99, except 55) → 5 points
 - Special case: 55 → 7 points
 - All others → 1 point

Setup

1. Shuffle the deck.
2. Deal 10 cards to each player.
3. Place 4 cards face-up on the table → they start 4 rows.

Play

Each round has 10 turns.

1. **Choose a Card** – Each player secretly picks one card from their hand and places it face-down.
2. **Reveal & Place** – All chosen cards are revealed at the same time and placed in ascending order:
 - **Rule 1:** Place your card in the row where the last card is the largest number that is still smaller than your card.
 - **Rule 2:** If more than one row fits, place it in the row with the smallest difference.
3. **6 nimmt! Rule** – If your card becomes the 6th card in a row, you must take the first 5 cards as penalty. Your card becomes the new row starter.
4. **Low Card Rule** – If your card is smaller than the last card of all rows, you must choose a row to pick up. Your card starts that row.

Scoring

- After all 10 turns, players count the bull points in their penalty piles.
- The game continues until someone reaches 66 or more bull points.
- The player with the lowest total wins.

2. Game Implementation and Features

This project will deliver a multiplayer version of 6 nimmt! developed using the Godot Engine, supporting both human and AI players.

Core Game Engine

- Full implementation of 6 nimmt! game.
- Handles game state, player hands, card placement, row management, and scoring.
- Supports 2–10 players: Any mix of human players (manual input) and AI opponents.

Graphics and User Interface (UI)

- Godot UI System with clean, user-friendly layouts.
- Each card shows its number and bull point value

Player Interaction

- **Input Options:**
 - Mouse input for selecting and playing cards.
- **Multiplayer Modes:**
 - Local pass-and-play (multiple players on same device).

AI Opponent Logic

- **Heuristic-based AI** with:
 - **Safe Play Heuristic** – prioritizes avoiding taking a row.
 - **Damage Control Heuristic** – if forced to take a row, minimizes bull points.
 - **Low Card Rule Handling** – chooses the row with the lowest bull points when forced.
- **Minimax Algorithm**
 - Only for 1 human and 1 AI player.
 - Alpha-Beta pruning implemented.

Extra Features (Enhancements)

- Game Over Screen showing scores, bull point totals, and declaring the winner.

LLM Usage

- For image generations



- Error Fixing (Async Await Issue)

GDScript Reference:

[GDScript reference — Godot Engine \(stable\) documentation in English](#)

Godot Engine Tutorial:

https://youtu.be/5V9f3MT86M8?si=98aC8m161_KXUXAD

Pillow (PIL) Library Reference:

[Reference - Pillow \(PIL Fork\) 11.3.0 documentation](#)